

Pre-training Question Embeddings for Improving Knowledge Tracing with Self-supervised Bi-graph Co-contrastive Learning

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Learning high-quality vector representations (aka. embeddings) of educational questions lies at the core of knowledge tracing (KT), which defines a task of estimating students' knowledge states by predicting the probability that they correctly answer questions. Although existing KT efforts have leveraged question information to achieve remarkable improvements, most of them learn question embeddings by following the supervised learning paradigm. In this paper, we propose a novel question embedding pre-training method for improving knowledge tracing with self-supervised *Bi*-graph *Co*-contrastive learning (*BiCo*). Technically, on the basis of self-supervised learning paradigm, we first select two similar but distinct views (i.e., representing objective and subjective semantic perspectives) as the semantic source of question embeddings. Then, we design a primary task (structure recovery) together with two auxiliary tasks (question difficulty recovery and contrastive learning) to further enhance the representativeness of questions. Finally, extensive experiments conducted on two real-world datasets show BiCo has a higher expressive power that enables KT methods to effectively predict students' performances.

CCS Concepts: • **Computing methodologies** → **Cognitive science**.

Additional Key Words and Phrases: Knowledge Tracing, Self-supervised Learning, Bi-graph, Co-contrastive Learning

1 INTRODUCTION

The recent COVID-19 epidemic has triggered the outbreak and proliferation of online education platforms which create an open as well as convenient environment for both students and teachers to learn and teach at home [1]. Now more than ever, it has become paramount to push the limits of online learning in every manner to keep flourishing the education system, where an essential component of online education is knowledge tracing (KT) aiming at modeling student's knowledge proficiency on the basis of their answers to a sequence of questions referred as interactions or responses [2].

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Specifically, the purpose of KT is twofold: (1) predicting the student's response score on next candidate question; (2) estimating the knowledge levels (i.e., knowledge states) of how the student masters on concepts. To achieve the goals of KT, question embeddings with high qualities play a crucial role, taking effect in inferring students' knowledge states for the final score prediction. Previous studies [3–9] have attempted to capture representations of knowledge states of students effectively by training rational question embeddings, which have yielded immense success on the field of KT. From such an inference perspective, an important premise of realizing KT is that all questions to be the input of KT models ought to be well represented in advance. Fig. 1 illustrates an instance of three educational questions to be represented in KT task.

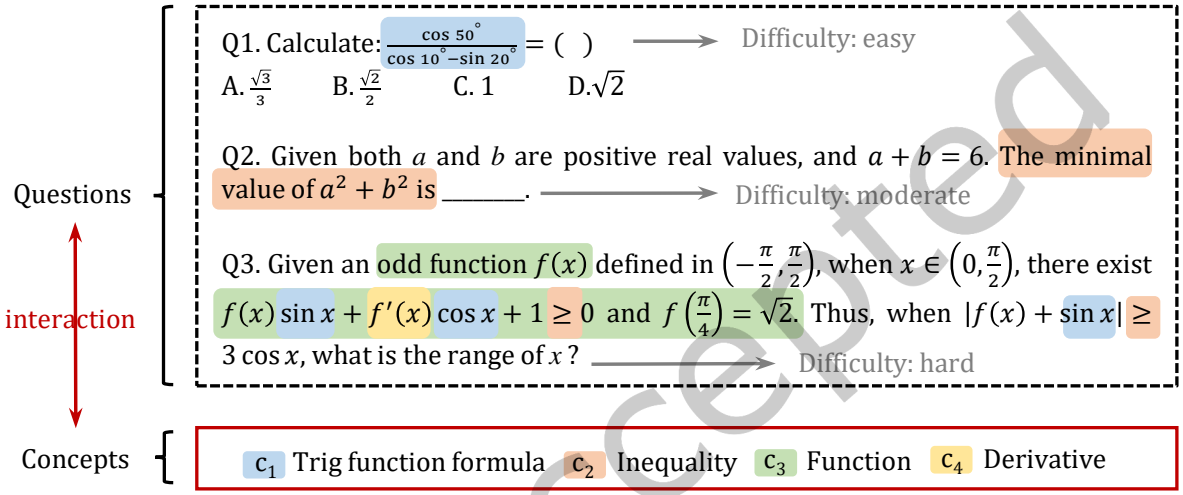


Fig. 1. An instance of three educational questions. Contents in the dotted-line box are text information of questions, where the gray fonts indicate difficulty information of them respectively. Rounded rectangle boxes with different colors in the dark red solid-line box represent four concepts related to these questions.

Despite the demanding usage of question embeddings of these mainstream KT models, fundamentally, the generation of question embeddings relies on at least one of the following rules (as shown in Fig. 2): (1) the question index and the corresponding response [4, 5, 7, 9]; (2) the question-concept relations [8, 10, 11]; (3) the text content of the question [6, 12–14]. For the first rule, DKT [7] is the first method proposing this way of representing a question. Specifically, DKT and most existing methods like DKVMN [9], SAKT [5], GKT [4] and so on, set the question embedding to be a one-hot encoding of the student interaction tuple that represents the combination of which exercise was answered and if the exercise was answered correctly or not. For the second rule, the main idea to represent a question is that concepts are the backbone embedded in questions so that a well-designed question embedding ought to contain its related concept information. There are a heaps of strategies to utilize the question-concept relations. For instance, GIKT [8] re-constructs the question-concept graph to learn the question node embeddings. CQKT [11] presents three types of partial order relations to calibrate Q-matrix (i.e., it contains question-concept relations) and LSTMCQ [10] also calibrate Q-matrix in a contextual way. Although CQKT and LSTMCQ use different strategies to calibrate Q-matrix, both methods learn the question embeddings by using question-concept relations with a similar manner. As for the third rule, those methods such as RKT [6], EKT [13], QuesNet [14] and DisenQNet [12] leverage text contents of questions as side information to represent questions.

Although the successful application of the above rules has demonstrated their effectiveness, these embeddings still suffer from the problem of data sparsity due to very limited student-question interactions and manually

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annotated sparse question-concept relations. Hence, the question embeddings learned from the sparse interaction space tend to be readily biased or over-fitted, which exerts a negative impact on the accuracy of inferring latent knowledge states. Recently, contrastive learning, as a special branch of self-supervised learning, becomes one of the most promising and suitable alternatives for compensating for the issue of data sparsity [15]. Albeit the common use of contrastive learning in diverse domains, including computer vision [16, 17], natural language processing [18, 19], and recommender systems [20, 21], the exploration of representing educational questions has received relatively less investigation.

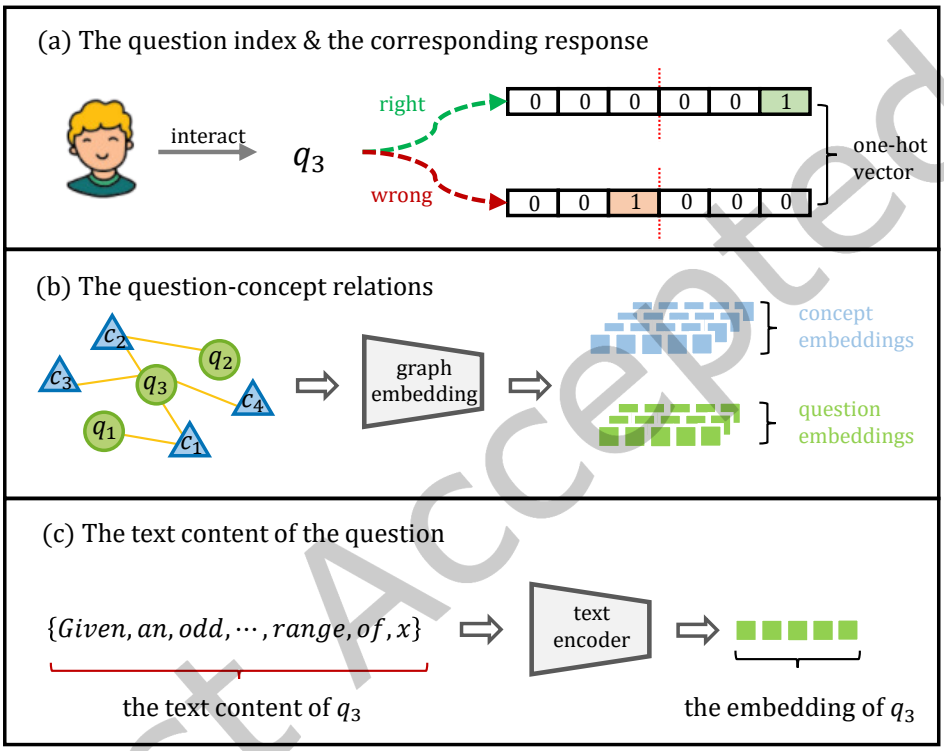


Fig. 2. Three mainstream rules of the generation of question embeddings.

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In effect, difficulties lie ahead to pre-train or represent educational question embeddings with contrastive learning, thus we need to carefully consider the unique characteristic of contrastive learning and the discrete nature of educational questions. This requires us to account for the following fundamental problems: (1) *How to select proper views for question embeddings?* Cross-view contrastive learning is desired for pre-training question embeddings. Despite various disparate views hidden in the student-question-concept interaction data, one fundamental demand is that the selected views should cover both objective and subjective semantics to provide internal and external clues. To be specific, from the perspective of objective semantics, the reliable relations between questions and concepts annotated by domain experts contains the objective as well as heterogeneous information, which can offer internal clues effectively. From the perspective of subjective semantics, subjective relations between questions can be derived from students' response logs so that it can provide external clues. (2) *How to keep contrastive difficult as well as task-relevant?* As is well recognized, an appropriate contrastive

task will further foster learning more discriminative embeddings [16, 17, 22]. If two views bear many similarities to each other, the mined signals will be too weak to make contributions to learning high-quality embeddings. Thereby, intuitively, we need to enforce these selected views to be more dissimilar but complementary when performing contrastive learning. Along with this, to devise a contrastive learning model for the specific task (i.e., pre-training question embeddings), it is necessary to make contrastive loss entail as much task-relevant information as possible.

Hinged on the above insights, in this paper, we strive to propose a novel question embeddings pre-training method for improving knowledge tracing with self-supervised **Bi**-graph **Co**-contrastive learning, dubbed BiCo. Specifically, we first exploit the 'question-to-concept' (Q2C) graph and the 'question-to-question' (Q2Q) graph to extract two different views that exhibit objective and subjective semantic connectivities of questions. Moreover, two distinct graph encoders over these two views are built, recursively leveraging different connectivity information to generate ground-truth samples by pulling similar samples together and pushing dissimilar ones apart in a co-training manner. Beyond recovering prior signals offered by edges in the Q2C graph, we obtain the final question embeddings with difficulty information by integrating feature vectors of questions. The contributions of our work are summarized as follows:

- We select two rational views that are able to cover both objective and subjective semantics to provide internal and external clues to our model.
- We devise a contrastive learning loss for the specific task (i.e., pre-training question embeddings) and keep contrastive difficult as well as task-relevant.
- Extensive experiments on two real-world datasets validate the effectiveness and superiority of BiCo, including quantitative comparison, qualitative analysis and case study.

The rest of this paper is organized as follows: We briefly review related work in Section 2. Some preliminaries are specified in Section 3. In Section 4, the proposed BiCo method is presented and discussed in detail. We conduct extensive experiments on two real-world datasets and comprehensively explore these results in Section 5. Section 6 concludes the whole paper and identifies future work which is likely to be carried out on the basis of the new findings exhibited in this paper.

2 RELATED WORK

2.1 Knowledge Tracing

The developed pre-training method is designed for knowledge tracing, a fundamental task, indirectly inferring the probability of students' success or failure on a candidate question based on sequences of their past attempts. Following the idea of mainstream KT methods, these KT methods can be divided into two broad strands of work: state-oriented methods and prediction-oriented methods.

On one hand, the state-oriented methods maintain vectors inferred by question embeddings to represent students' knowledge states, such as DKT [7], DKVMN [9], AKT [3], GKT [4], and GIKT [8]. As one of the representative methods, DKT [7] represents students' knowledge states with the hidden states of LSTM, and predicts students' responses by feeding the knowledge states to a binary classifier. As students' cognition improves when they interact with the questions, DKT updates their knowledge states by using question representation and students' responses. Many works obtain a better performance by extending DKT. For instance, DKVMN [9] stores the concept representation in the key matrix and the knowledge states in the value matrix. AKT [3] couples flexible attention-based neural network models with a series of novel, interpretable model components. GKT [4] builds a skill relation graph and learns their relation explicitly. Similarly, GIKT [8] proposes a framework to employ high-order question-skill relation graphs into question and skill representations for knowledge tracing.

On the other hand, the prediction-oriented methods attempt to incorporate the similarities between question embeddings with students' forgetting rate to simulate the behaviors of students for generating the final score

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prediction in a straightforward manner, like SAKT [5] and RKT [6]. SAKT [5] is the first method using attention mechanism to capture long term dependencies between student-question interactions. Building upon this idea, RKT [6] models a student's interaction history and predicts her performance on the next exercise by considering contextual information obtained from its relation with the past exercises and the forget behavior of the student. Both methods utilize attention mechanism to replace the RNNs to assess students' knowledge states, while the estimated knowledge states are not interpretable.

Summary. In short, an important premise of realizing KT is that all questions to be input ought to be well represented in advance. Although these aforementioned methods have made great progress, none of them can bypass the inherent sparsity problems on KT datasets and their generated question embeddings are not flexible. For this, in this paper, we attempt to develop a pre-trained question embedding method to compensate for the above issues.

2.2 Question Embedding

Recently proposed question embedding methods aim to be model-agnostic for making question embeddings flexible without weakening their representative ability. QuesNet [14] is the first attempt to pre-train question embeddings from the KT datasets with high heterogeneity. In addition, DisenQNet [12] is a novel unsupervised model, which aims to discriminate the diverse characteristics of questions and learn the disentangled question embeddings. The two model-agnostic methods encode the questions' text contents into question embeddings to capture semantic characteristics of questions. Nevertheless, the requirement that the text descriptions of questions have to be known beforehand is actually frequently violated [24], because most of these contents are private. In order to get rid of the limitation of texts, inspired by LINE [23], PEBG [24] is designed to extract and exploit plentiful latent information between questions and concepts. Although PEBG models question embeddings from question-concept interaction graph without text information, it introduces a number of extra embarrassingly simple folded signals between questions (concepts) and produces noises [25]. Therefore, in this paper, we strive to model question embeddings which have the ability of get rid of the limitation of insufficient labels (data sparsity issue).

Summary. In the face of aforementioned challenges, a primary goal is to build pre-training question embedding model which should abandon text contents of questions, but is nevertheless effective. Therefore, in this paper, we strive to model question embeddings which have the ability of get rid of the limitation of insufficient labels.

2.3 Contrastive Learning and Multi-view Graph Representation

In recent years, self-supervised learning [15] is argued to be an antidote to the data sparsity issue, which can uncover ground-truth samples from the training data and supervise models using the generated labels. The methods utilizing contrastive learning obtain embeddings by contrasting positive examples against negative ones and yield immense success including computer vision [16, 17], natural language processing [18, 19], and recommender systems [20, 21]. We predominantly focus on recapping the methods related to graph contrastive learning. Concretely, DGI [26] constructs local patches and global summary as positive pairs to make them the ground truth of each other. Along with this, GMI [27] is developed to contrast the center node against the local ones from their features and topological structure. Furthermore, in the field of heterogeneous graphs, DMGI [28] contrasts the original graph against the corrupted graph on every single view, and meta-path, designing a consensus regularization to guide the aggregation of different meta-paths. Inspired by recent advances in contrastive learning in the field of graph mining, we strive to design a novel question embedding pre-training method for addressing the aforementioned issues.

Furthermore, learning representations from multi-view data can achieve strong generalization performance. A typical process of multi-view representation learning is constituted by three major steps: (i) identify multiple

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views that can sufficiently represent the data; (ii) carry out representation learning on each view; and (iii) combine the multiple view-specific representations. MultiKE [28] is an entity alignment framework based on multi-view knowledge graph (KG) embedding, and divide the various features of KGs into multiple subsets (called views), which are complementary to each other. Besides, JOIE [29] employs both cross-view and intra-view modeling that learn on multiple facets of the knowledge base (KB), which jointly encodes both the ontology and instance views of the KB. To further improve contrastive representation learning on node and graph classification tasks, Hassani and Khasahmadi [30] introduce a self-supervised approach for learning node and graph level representations by contrasting structural views of graphs. They show that unlike visual representation learning, increasing the number of views to more than two or contrasting multi-scale encodings do not improve performance, and the best performance is achieved by contrasting encodings from first-order neighbors and a graph diffusion.

Summary. Inspired by recent advances in contrastive learning and multi-view graph representation, an intuition is that multi-view or even cross view are significant for graph contrastive learning. Therefore, we strive to apply graph contrastive learning with multi-view or even cross view for mining question-concept graph to learn question embeddings.

3 PRELIMINARIES

In this section, we introduce some preliminary knowledge about question embedding, including necessary definitions and notations used in this paper.

Table 1. The key mathematical notations and descriptions

Notation	Description
N, M, K	the total number of students; the total number of questions; the total number of concepts
$\mathcal{X}_u$	student u 's history interaction records, where $u \in [1, N]$
$\mathcal{S}, \mathcal{Q}, \mathcal{C}$	the set of students, questions and concepts
$\mathcal{G}_{q2c}, \mathcal{G}_{q2q}$	the Q2C graph and the Q2C graph
q_i	an arbitrary question i
d_i	the difficulty of question i
$\mathbf{q}_i, \mathbf{c}_k$	the embedding of question i and the embedding of concept k
$\mathbf{Q}, \mathbf{C}$	the question embedding matrix and the concept embedding matrix to be learned
$\mathbf{X}_{obj}, \mathbf{X}_{sub}$	embedding matrices under the objective view and the subjective view
$\bar{\mathbf{q}}_i, \widehat{\mathbf{q}}_i$	two embeddings of question i under the objective view and the subjective view, respectively

Let $\mathcal{S} = \{s_1, s_2, ..., s_N\}$ be the set of N students, $\mathcal{Q} = \{q_1, q_2, ...q_M\}$ be the set of M questions and $\mathcal{C} = \{c_1, c_2, ...c_K\}$ be the set of K concepts. In KT task, given a student u 's history interaction records $\mathcal{X}_u = (x_1^u, ..., x_t^u)$ where $x_t^u = (q_t, r_t)$ and $r_t \in \{0, 1\}$ is the response that the student responds to a question correctly ($r_t = 1$) or incorrectly ($r_t = 0$) at the time step t. Given the student u's history interaction records $\mathcal{X}_u$ and the next question q_t, KT aims to estimate the probability of answering q_{t+1} correctly: $\hat{r}_{t+1} = p(r_{t+1} = 1|\mathcal{X}_u, q_{t+1})$. Here we predominantly introduce the information we will leverage to train question embeddings in our model, including the graph information and the difficulty information. For better description, the key notations are summarized in Table 1.

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Definition 3.1. (Question-to-concept graph) The question-to-concept graph (i.e., Q2C graph), denoted as $\mathcal{G}_{q2c} = (\mathcal{Q} \cup \mathcal{C}, \mathcal{R}_{q2c})$, is an undirected graph, where $\mathcal{R}_{q2c} = \{r_{q_i \leftrightarrow c_k} | q_i \in \mathcal{Q}, c_k \in \mathcal{C}, r_{q_i \leftrightarrow c_k} \in \{0, 1\}\}$, and $r_{q_i \leftrightarrow c_k} = 1$ indicates that the question q_i involves the concept c_k, otherwise $r_{q_i \leftrightarrow c_k} = 0$.

Definition 3.2. (Question-to-question graph) The question-to-question graph (i.e., Q2Q graph), denoted as $\mathcal{G}_{q2q} = (\mathcal{Q}, \mathcal{R}_{q2q})$, is an undirected graph, where $\mathcal{R}_{q2q} = \{r_{q_i \leftrightarrow q_j} | q_i, q_j \in \mathcal{Q}, r_{q_i \leftrightarrow q_j} \in \{0, 1\}\}$, and $r_{q_i \leftrightarrow q_j} = 1$ suggests that there exists a semantic relation between the question q_i and q_j, otherwise $r_{q_i \leftrightarrow q_j} = 0$.[1]

Definition 3.3. (Question difficulty) Given a question q_i, the ground-truth question difficulty d_i is defined as the slipping ratio, computing as follows:

$$d_i = \frac{\sum_{s_u \in S} \sum_{x_j^u \in X_u} \mathbb{I}[(q_j = q_i) \wedge (r_i = 0)]}{\sum_{s_u \in S} \sum_{x_j^u \in X_u} \mathbb{I}(q_j = q_i)} \tag{1}$$

where $\mathbb{I}(\cdot)$ denotes the indicator function. d_i with a bigger value means that the question q_i is more difficult since more students answer this question incorrectly, and vice versa.

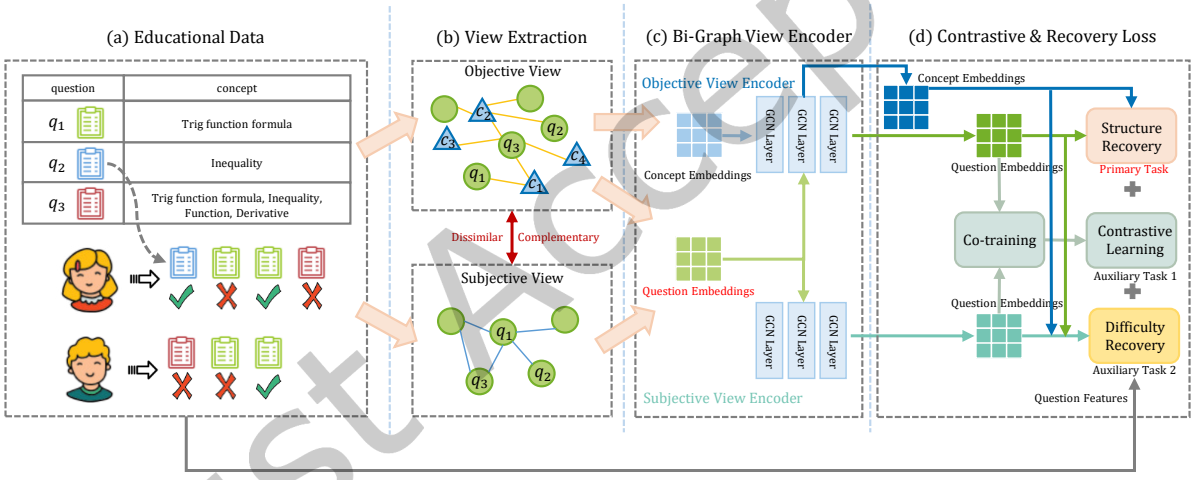


Fig. 3. The overview framework of BiCo: (a) The objective relations between questions and concepts; (b) The extracted objective and subjective views from the educational data; (c) The objective view encoder and subjective view encoder with GCN layers; (d) The primary task together with two auxiliary tasks. Best viewed in color.

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4 METHODOLOGY

Inspired by recent advances in contrastive learning in the field of machine learning with graphs, a multitude of studies have manifested the fact that transplanting contrastive learning frameworks on graph data also performs well [21]. Therefore, in this section, we strive to devise a novel question embedding pre-training method for improving knowledge tracing with self-supervised Bi-graph Co-contrastive learning (BiCo), whose

[1]Following [25], the value of $r_{q_i \leftrightarrow q_j}$ is determined by computing *NMI* (Normalized Mutual Information) between question q_i and q_j based on students' response logs, so the computation details are omitted.

overview framework is shown in Fig. 3. Specifically, we first exploit the 'question-to-concept' (Q2C) graph and the 'question-to-question' (Q2Q) graph to extract two rational views that exhibit objective and subjective semantic connectivities of questions. Moreover, two distinct graph encoders over these two views are built, recursively leveraging different connectivity information to generate ground-truth samples by pulling similar samples together and pushing dissimilar ones apart in a co-training manner. Finally, we devise a contrastive learning loss for the specific task (i.e., pre-training question embeddings) and keep contrastive difficult as well as task-relevant.

4.1 View Extraction

As discussed in the Section 1, it is of great necessity to select rational and proper views covering objective and subjective semantics for question embeddings. Along this line, in this subsection, we aim to extract two distinct views which are Q2C graph-guided objective view and Q2Q graph-guided subjective view from the training data.

Since the Q2C graph which is typically annotated by educational experts is the source of information about questions and concepts, it plays a pivotal role in realizing KT and cognitive diagnosis (CD) [11]. Moreover, the Q2C graph can provide objective and reliable relations between questions and concepts to question embeddings that are explicitly required in the fields of KT and CD. Thereby, the selected objective view is rooted in Q2C graph, which can help our model to generate question embeddings approaching ground-truth questions.

Along with this, conventionally, a substantial amount of students' response logs are overlooked when constructing question embeddings [29]. However, with the help of response logs, we have the ability to reveal external relations between questions by taking right or wrong answers that students left into consideration in an indirect manner [6, 30]. Thus, we utilize *NMI* to measure the association $\tilde{r}_{q_i \leftrightarrow q_j}$ between any two questions (q_i and q_j) depending on response logs. Subsequently, to construct Q2Q graph, we try to build an edge between any two questions by setting a sparsity factor λ:

$$r_{q_i \leftrightarrow q_j} = \begin{cases} 1, \tilde{r}_{q_i \leftrightarrow q_j} \geq \lambda \\ 0, \text{otherwise} \end{cases} \tag{2}$$

Therefore, the Q2C graph-guided objective view and Q2Q graph-guided subjective view are extracted from the training data in an explicit and implicit way.

4.2 Learning Bi-Graph Encoders over Extracted Views

In the subsection 4.1, we endow two types of aforementioned graphs with specific semantics (i.e., objective and subjective) which take effects in the self-supervised graph co-training scenario. Although our goal is to devise a model-agnostic pre-training question embedding framework that can drive a multitude of downstream KT models, for the requirement sufficing for capability of the proposed self-supervised bi-graph co-training, we need to construct the bi-graph encoder with graph convolutions on two extracted views to capture high-level connectivities.

Specifically, we first initialize question and concept embeddings by two d-dimensional trainable matrices $\mathbf{Q} \in \mathbb{R}^{M \times d}$ and $\mathbf{C} \in \mathbb{R}^{K \times d}$. For question q_i and the concept c_k, they are initialized by multiplying the corresponding one-hot representation vectors and the corresponding trainable matrix, respectively:

$$\begin{cases} \mathbf{q}_i = \tilde{\mathbf{q}}_i^T \mathbf{Q} \\ \mathbf{c}_k = \tilde{\mathbf{c}}_k^T \mathbf{C} \end{cases} \tag{3}$$

where $\mathbf{q}_i, \mathbf{c}_k \in \mathbb{R}^d$, $\tilde{\mathbf{q}}_i \in \{0, 1\}^M$ and $\tilde{\mathbf{c}}_k \in \{0, 1\}^K$.

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Due to the effectiveness and lightweight architecture of LightGCN [34], we first adopt its message propagation mechanism to encode collaborative signals from Q2C graph as below:

$$\mathbf{q}_i^{(l+1)} = \sum_{k \in N_q^{obj}} \frac{\mathbf{c}_k^{(l)}}{\sqrt{|N_q^{obj}||N_c^{obj}|}} \tag{4}$$

$$\mathbf{c}_k^{(l+1)} = \sum_{i \in N_c^{obj}} \frac{\mathbf{q}_i^{(l)}}{\sqrt{|N_c^{obj}||N_q^{obj}|}} \tag{5}$$

where $\mathbf{q}_i^{(l)}$ and $\mathbf{c}_k^{(l)}$ represent the encoded embeddings of question q_i and concept c_k under the l-th graph convolution layer. N_q^{obj} and N_c^{obj} represent the set of concepts covered by q_i and questions related to c_k under the objective view, respectively. After the message propagation of L layers, we further combine embeddings obtained at each layer to form the updated representation of a question (a concept):

$$\bar{\mathbf{q}}_i = \sum_{l=0}^{L} \alpha_l \mathbf{q}_i^{(l)}; \bar{\mathbf{c}}_k = \sum_{l=0}^{L} \alpha_l \mathbf{c}_k^{(l)} \tag{6}$$

where $\alpha_l > 0$ indicates the importance of the l-th layer embeddings in composing updated embeddings, and it can be treated as a hyperparameter to be tuned manually. In our experiments, we find that setting α_l as $1/(l+1)$ leads to good performance in general. In particular, $\mathbf{q}_i$ and $\mathbf{c}_k$ are encoded embeddings of q_i and c_k under the 0-th graph convolution layer (i.e., $\mathbf{q}_i^{(0)} = \mathbf{q}_i$, $\mathbf{c}_k^{(0)} = \mathbf{c}_k$). Hence, we can obtain updated question and concept embeddings to compose the embedding matrix $\mathbf{X}_{obj}$ through objective view encoding:

$$\begin{aligned} \mathbf{X}_{obj} &= \left[\mathbf{X}_{obj}^{q}, \mathbf{X}_{obj}^{c}\right]^T \\ &= [\bar{\mathbf{q}}_1, ..., \bar{\mathbf{q}}_M, \bar{\mathbf{c}}_1, ..., \bar{\mathbf{c}}_K]^T \end{aligned} \tag{7}$$

In contrast, the subjective view depicts relations between questions from the external perspective. Likewise, the subjective view encoder also conducts LightGCN on Q2Q graph. Thereby, through the subjective view encoding, we can obtain another type of updated question embeddings from the embedding matrix $\mathbf{X}_{sub} = [\widehat{\mathbf{q}}_1, ..., \widehat{\mathbf{q}}_M]^T$.

In short, whether the objective view encoder or subjective view encoder, the parameters to be learned are initialized question embeddings (i.e., $\mathbf{q}_1, ..., \mathbf{q}_M$) and concept embeddings (i.e., $\mathbf{c}_1, ..., \mathbf{c}_K$).

4.3 Self-supervised Bi-graph Co-contrastive Learning

4.3.1 Excavating Self-Supervision Signals by Bi-Graph Co-Training. Here, we will present the principle in which bi-graph co-training excavates self-supervision signals assisting to generate and tune finer-grained question embeddings. As mentioned before, we construct the bi-graph view encoder over two distinct views that have the ability to offer complementary information to each other. As such, refining each encoder by mining information from the other view is natural. This can be realized by the follow-on co-training mechanism.

Specifically, given a question q_i in the subjective view, we obtain its positive and negative concept samples by using its corresponding representation learned over the objective view:

$$\begin{cases} \mathbf{y}_{obj}^{q_i} = \text{softmax}\left(\mathbf{score}_{obj}^{q_i}\right) \\ \mathbf{score}_{obj}^{q_i} = \mathbf{X}_{obj}^{c} \bar{\mathbf{q}}_i \end{cases} \tag{8}$$

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where $\mathbf{y}_{obj}^{q_i} \in \mathbb{R}^K$ indicates the computed likelihood of each concept being related to question q_i in the objective view. Here, the softmax function takes as input a vector $\mathbf{y}_{obj}^{q_i}$ of K (i.e., the total number of concepts) real numbers, and normalizes it into a probability distribution consisting of K probabilities proportional to the exponentials of the input numbers. Moreover, $\tilde{\mathbf{q}}_i$ can be considered as a linear classifier, and $\mathbf{X}_{obj}^{c}$ can be considered as the unlabeled sample set.

With the premise of obtaining $\mathbf{y}_{obj}^{q}$, those concepts with top-k highest confidence can be selected as positive samples acting as the semantically related ground truths to supervise the subjective encoder. For this, the strategy of sampling positives is defined as $c_{sub}^{q^+} = \text{top-}k\left(\mathbf{y}_{obj}^{q}\right)$. In fact, as for the way to select negative samples, a straightforward idea is to take the concepts with the lowest scores. In our previous investigation, such a way can only choose easy samples (i.e., concepts) which contribute little. Unfortunately, such a solution can solely pick too easy samples to make little contributions to co-training. In contrast, we attempt to select k negatives from the concepts ranked in the top 10% in $\mathbf{y}_{obj}^{q}$ except for the positive samples randomly, as augmented negative samples to build $c_{sub}^{q^-}$. These selected concepts can be treated as hard negative samples providing more informative signals, and are not likely to fall into the set of false concepts misleading the co-training.

Similarly, we utilize the analogous manner to filter valuable positive and negative samples for the subjective view encoder:

$$\begin{cases} \mathbf{y}_{sub}^{q_i} = \text{softmax}\left(\mathbf{score}_{sub}^{q_i}\right) \\ \mathbf{score}_{sub}^{q_i} = \mathbf{C}\widehat{\mathbf{q}}_i \end{cases} \tag{9}$$

Eq. (9) mainly differs from Eq. (8) in which when selecting positive samples, $\mathbf{C}$ is leveraged rather than $\mathbf{X}_{obj}^{c}$ as the subjective view encoder does not output convolved concept embeddings. Specifically, $\mathbf{score}_{sub}^{q_i} = \mathbf{C}\widehat{\mathbf{q}}_i$ aims to compute the relations between the embedding of question q_i under the subjective view and every single concept embedding in $\mathbf{C}$. Thereafter, we use the softmax function to normalize $\mathbf{y}_{sub}^{q_i}$ into a probability distribution.

In each batch of the training procedure, whether positive or negative pseudo-labels for each question in every single view are reconstructed iteratively, and then are delivered to the other view as the underlying related concepts for distilling question and concept embeddings. Such a cross-scale process reflects our intuition rooting that the concept samples with a high confidence score to be the concept related to the question q_i in one view is also supposed to be needed in the other view. Iterating the process is expected to generate more informative and harder examples that have the capacity to provide numerous clues to learn question embeddings. Conversely, the bi-graph encoder can evolve with the supervised signals offered by these informative examples as well, distilling examples with high-quality and useful recursively.

4.3.2 Contrastive Learning. By using these pseudo-labels, the well-designed task-relevant contrastive task can be conducted for refining encoders. In the pre-training question embedding scenarios, we hypothesize that the question is the most related to the concept where there exists an edge as the ground truth between them in Q2C graph. Thus, the similarity between question embeddings and selected concept samples can be maximized (minimized) in a cross-scale manner, together with the given question embeddings as the question context. One benefit of such a solution is that the picked positive samples can reflect relatively objective relations between questions and concepts well. Formally, given a question q_i and the generated ground-truths, we follow InfoNCE [35] and define a contrastive loss to learn informative and effective question embeddings by pulling semantically

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similar positive pairs together and pushing apart negative pairs:

$$L_{cl} = -\log_2 \frac{\sum_{k \in c_{sub}^{q^+}} \phi\left(\bar{\mathbf{q}}_i, \bar{\mathbf{c}}_k\right)}{\sum_{k \in c_{sub}^{q^+}} \phi\left(\bar{\mathbf{q}}_i, \bar{\mathbf{c}}_k\right) + \sum_{j \in c_{sub}^{q^-}} \phi\left(\bar{\mathbf{q}}_i, \bar{\mathbf{c}}_j\right)} - \log_2 \frac{\sum_{k \in c_{obj}^{q^+}} \phi\left(\widehat{\mathbf{q}}_i, \mathbf{c}_k\right)}{\sum_{k \in c_{obj}^{q^+}} \phi\left(\widehat{\mathbf{q}}_i, \mathbf{c}_k\right) + \sum_{j \in c_{obj}^{q^-}} \phi\left(\widehat{\mathbf{q}}_i, \mathbf{c}_j\right)} \tag{10}$$

where $\phi\left(x_1, x_2\right) = \exp\left[\operatorname{sim}\left(x_1, x_2\right)/\tau\right]$, τ denotes a temperature hyperparameter (we empirically use 0.2 in our experiments), and $\operatorname{sim}(\cdot)$ denotes the cosine similarity between two vectors.

4.4 Multi-task Learning

4.4.1 Structure Recovery. Actually, in Q2C graph, edges exist between question nodes and concept nodes, indicating explicit signals. Inspired by the approach of modeling 1st-order proximity in LINE [23], likewise, we also attempt to model explicit relations by estimating the local proximity between question and concept in the embedding space. In detail, the local proximity between q_i and c_k is modeled as:

$$\hat{r}_{q_i \leftrightarrow c_k} = \text{sigmoid}\left(\bar{\mathbf{q}}_i^T \bar{\mathbf{c}}_k\right) \tag{11}$$

Since the objective view can reflect question-concept connectivity in a finer-grained granularity, we choose to utilize encoder over this view as the main encoder to recover structure information in Q2C graph. Afterwards, to preserve structure relations, the local proximity is enforced to be close to question-concept relations in Q2C graph via a cross-entropy loss function:

$$L_{sr} = -\sum_{q_i \in Q} \sum_{c_k \in C} [r_{q_i \leftrightarrow c_k} \log_2 \hat{r}_{q_i \leftrightarrow c_k} + (1 - r_{q_i \leftrightarrow c_k}) \log_2 (1 - \hat{r}_{q_i \leftrightarrow c_k})] \tag{12}$$

4.4.2 Difficulty Recovery. One of the crucial goals of some downstream tasks (e.g., question recommendation or adaptive testing) is enabling questions to be suitably challenging, and it is a common notion that a good question should be neither too easy nor too difficult [36]. Thus, in our method, we attempt to make questions possess difficulty information, which reflects our intuition: difficulty information of questions plays a significant role in the KT task, which, however, is not contained in Q2C graph. Specifically, we leverage attribute features of questions interacting with question and concept embeddings to learn high quality question embeddings.

As suggested in [24], the feature vector of q_i can be represented as $\tilde{\mathbf{f}}_i$. Furthermore, for the target question q_i, we assume that the set of concepts related to q_i is N_{q_i}, we use the average representation of all concept embeddings in N_{q_i} as the related concept feature of q_i, denoted as:

$$\mathbf{q}_i^c = \frac{1}{\left|N_{q_i}\right|} \sum_{c_k \in N_{q_i}} \bar{\mathbf{c}}_k \tag{13}$$

Sequently, to recover objective and subjective difficulty information of the target question q_i effectively, we project concatenated vector $[\bar{\mathbf{q}}_i || \widehat{\mathbf{q}}_i || \mathbf{q}_i^c || \tilde{\mathbf{f}}_i]$ into the final target question embedding $\hat{\mathbf{q}}_i \in \mathbb{R}^d$ by using a feed-forward neural network with one hidden layer, and further leverage a non-linear layer to map $\hat{\mathbf{q}}_i$ to a difficulty approximation:

$$\hat{d}_i = \text{sigmoid}(\mathbf{w}_d^T \cdot \hat{\mathbf{q}}_i + b_d) \tag{14}$$

where $\mathbf{w}_d$ and b_d are trainable parameters.

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Finally, depending on the *Definition 3*, we design the following loss function to measure the difficulty approximation error:

$$L_{dr} = \sum_{i=1}^{M} \left(d_i - \hat{d}_i\right)^2 \tag{15}$$

4.4.3 Joint Optimization. Finally, in order to generate comprehensive question embeddings that preserve structure information, objective/subjective semantics and question difficulty information simultaneously, we unify the structure recovery task with the auxiliary SSL task and the difficulty recovery task to form a joint optimization framework, namely, we solve:

$$\min_{\theta} L = \min \left[L_{sr} + \gamma L_{cl} + (1-\gamma)\, L_{dr}\right] \tag{16}$$

where γ is a balance factor, and θ denotes the set of all the parameters to be learned in the BiCo framework, that is, the question embeddings, concept embeddings, and parameters of the feed-forward neural network. We optimize the proposed model via back propagation and learn the embeddings of questions and concepts. In the end, we can use them in downstream KT tasks like DKT, DKVMN, GIKT, et al.

4.5 BiCO Extension with Loss Enhancement

Inspired by [20], we can easily draw a conclusion that optimizing the CL loss (Eq. (10)) can be seen as an implicit way to debias because a more uniform representation distribution can preserve the intrinsic characteristics of question as well as concept embeddings and improve the generalization ability. For this, we attempt to extend the BiCo framework by enhancing the CL loss, denoted as BiCoEE.

Specifically, in the subsection 4.2, we conducts LightGCN on Q2C graph to obtain question and concept embeddings. However, considering that the Q2C graph is a bipartite graph, information propagation with GNN-based model for even times on the graph naturally aggregates information of homogeneous structural neighbors which makes it convenient to extract the potential neighbors within questions or concepts. Along this line, the embeddings of homogeneous neighborhoods from the even layer (e.g., 2, 4, 6) output of the GNN model can be obtained. With these question embeddings, it is efficient to model the relation between questions/concepts and their homogeneous neighbors. Thus, we view the question embedding themself and the embedding of the corresponding output of the even-numbered layer GNN as positive pairs to minimize the distance between them as follows:

$$L_{obj}^{Q} = -\sum_{i=1}^{M} \log_2 \frac{\exp((\mathbf{q}_i^{(p)} \cdot \mathbf{q}_i^{(0)})/\tau)}{\sum_{j=1}^{M} \exp((\mathbf{q}_i^{(p)} \cdot \mathbf{q}_j^{(0)})/\tau)} \tag{17}$$

where p is even number, and in our experiment, we find that $p = 4$ results in good performance in general. Similarly, the loss enhancement of concept side L_{obj}^{C} can be formulated as:

$$L_{obj}^{C} = -\sum_{k=1}^{K} \log_2 \frac{\exp((\mathbf{c}_k^{(p)} \cdot \mathbf{c}_k^{(0)})/\tau)}{\sum_{j=1}^{K} \exp((\mathbf{c}_k^{(p)} \cdot \mathbf{c}_j^{(0)})/\tau)} \tag{18}$$

Furthermore, to enhance the loss function over subjective view encoder, we exert a constraint on the subjective view encoder, which is actually minimizing the cosine similarity between different question embeddings:

$$L_{sub} = -\sum_{i=1}^{M} \log_2 \frac{\exp(1/\tau)}{\sum_{j=1}^{M} \exp(\mathbf{q}_i^T \mathbf{q}_j/\tau)} \tag{19}$$

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where it will push connected question nodes away from the high-degree hubs in the embedding space and lead to a more uniform distribution even under the influence of the primary loss. Among this, the Eq. (16) can be rewritten as follows:

$$\min_{\theta} = \min[L_{sr} + \gamma(L_{cl} + L^{Q}_{obj} + L^{C}_{obj} + L_{sub}) + (1-\gamma)L_{dr}] \tag{20}$$

5 EXPERIMENTS

In this section, we conduct extensive experiments to evaluate the proposed BiCo framework. To be specific, the following four research questions need to be answered:

- **RQ1**: Comparing with the state-of-the-art question embedding methods without text information (i.e., PEBG), how does our proposed BiCo method perform in terms of accuracy?
- **RQ2**: How about the KT models' performance when BiCo blend students' response logs into current models?
- **RQ3**: How do the hyperparameters tuned in the Section 4 affect BiCo effectiveness?
- **RQ4**: How about the interpretation of question difficulty recovered by BiCo method?

5.1 Datasets

Two real-world datasets have been utilized for our experiments. One is Assist09 (ASSISTments2009-2010 skill builder), an open dataset collected by the ASSISTments online tutoring systems, which is widely used by researchers for purposes such as KT and student modeling. Besides, we choose the public corrected version which mitigates the duplicated data problem proposed by previous research [30]. The other dataset is Junyi supplied by an E-learning platform established in 2012 on the basis of the open-source code released by Khan Academy, which is collected by JunyiAcademy in 2015 [6]. Before answering aforementioned research questions, it is of great necessity to process these two datasets for fitting our method better. Table 2 summarizes basic statistics of these two preprocessed datasets.

Table 2. The statistics of both datasets

Statistics	Assist09	Junyi
#Students	2,493	10,000
#Questions	17,746	835
#Concepts	123	835
#Response records	267,415	353,835
#Concepts per question	1.20	1.00
#Questions per concept	107.27	35.38

5.1.1 Experimental Setup. We set the embedding size $d = 64$ [24] and $L = 3$ (Eq. (6)) [34]. For those significant hyperparameters in BiCo, we select λ and γ from $(0, 1)$ for both datasets and select k from $\{3, 5, 7, 9, 10, 11, 15, 20, 25\}$ [37]. Besides, we implement the initialization of the whole parameters by *Xavier*. Meanwhile, we train the BiCo with the *Adam* optimizer. We choose learning rate within the range of $[1e-4, 5e-3]$, and tune patience for early stopping within the range of $[5, 50]$ [22]. As for the dropout rate, we test ranging from 0.1 to 0.5 with step 0.05, and we use the default mini-batch size of 64. Moreover, we use Pytorch to implement our method and conduct our experiments on a server with NVIDIA Tesla V100S GPU with 32GB of video memory.

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Table 3. Comparison results (%±σ) on student performance prediction on Assist09.

Train/Test ratio		50%/50%			60%/40%			70%/30%			80%/20%		
Methods		RMSE	ACC	AUC	RMSE	ACC	AUC	RMSE	ACC	AUC	RMSE	ACC	AUC
Baseline	DKT	47.12±4.7	63.17±5.8	68.31±5.7	46.74±6.8	63.89±8.1	68.74±5.6	46.43±4.2	71.42±2.1	73.73±3.2	45.88±3.1	73.12±2.8	74.07±3.0
	DKVMN	46.83±3.1	64.84±3.2	69.28±2.9	45.31±4.3	64.73±2.5	69.44±4.6	45.71±4.3	72.18±3.2	74.61±2.8	44.14±3.2	74.75±3.0	75.33±2.5
	SAKT	46.30±0.9	65.19±2.9	70.85±1.6	44.97±2.9	65.92±3.2	70.26±3.1	44.17±2.8	73.45±2.2	75.14±1.8	43.52±2.4	75.82±2.2	76.27±2.3
	AKT-NR	45.87±1.5	66.74±3.3	71.82±1.1	44.26±1.8	66.41±2.1	71.61±2.0	43.81±1.7	74.24±2.6	76.87±2.5	42.87±1.7	76.44±0.9	77.82±2.2
	GIKT	44.11±0.9	67.21±2.1	72.14±1.8	43.64±0.7	67.85±0.9	72.27±2.7	42.95±0.8	75.92±1.5	77.64±2.1	41.13±2.4	77.17±1.8	79.44±1.4
Pre-training based	PEBG+DKT	45.27±2.1	65.80±1.6	70.99±3.2	44.11±1.2	67.36±1.1	71.15±0.9	43.54±2.2	74.70±2.1	76.46±1.9	42.41±2.1	76.81±1.4	78.19±2.3
	PEBG+DKVMN	44.64±1.8	66.23±2.4	71.52±1.3	43.83±0.9	67.99±2.3	71.92±1.6	42.76±1.8	75.83±1.3	77.87±2.1	41.25±1.6	77.47±2.3	79.75±1.1
	BiCo+DKT	42.13±3.2	68.33±1.1	71.84±2.7	41.87±1.0	69.11±2.6	71.50±1.3	41.35±0.9	75.41±1.7	76.20±1.5	40.17±0.8	77.32±0.7	78.68±0.9
	BiCo+DKVMN	41.15±1.9	72.14±0.3	73.11±0.9	40.63±1.3	72.47±1.2	73.37±2.0	39.84±2.2	77.19±0.8	78.31±1.6	38.93±1.4	78.87±1.2	80.16±1.8
	BiCo+SAKT	41.93±0.7	71.25±0.8	72.17±0.3	41.31±0.8	70.91±1.4	72.42±1.8	40.73±1.4	75.85±1.3	77.27±2.1	39.62±2.3	77.61±1.8	78.91±2.1
	BiCo+AKT-NR	41.27±0.5	71.81±2.1	72.52±1.5	41.14±1.2	71.51±1.6	72.81±0.9	40.21±1.9	76.22±2.1	77.89±0.8	39.24±1.3	78.19±1.7	79.34±1.2
	BiCo+GIKT	39.83±1.5	72.92±1.2	73.90±0.8	40.18+0.7	73.22±1.3	74.18±1.2	39.24±0.8	77.92±1.1	79.18±0.7	38.20±1.1	79.24±0.9	81.16±0.4
	BiCoEE+GIKT	**39.14±0.2**	**73.4±1.1**	**74.27±0.6**	**39.76±0.9**	**73.92±1.1**	**74.63±1.0**	**38.43±0.9**	**78.32±0.6**	**79.94±0.2**	**37.41±0.8**	**80.1±0.3**	**81.63±0.2**

Table 4. Comparison results (%±σ) on student performance prediction on Junyi.

Train/Test ratio		50%/50%			60%/40%			70%/30%			80%/20%		
Methods		RMSE	ACC	AUC	RMSE	ACC	AUC	RMSE	ACC	AUC	RMSE	ACC	AUC
Baseline	DKT	49.62±4.6	59.91±5.1	66.31±5.8	49.37±4.6	62.74±5.1	66.17±5.8	46.74±4.2	69.42±5.1	70.30±4.8	46.22±3.9	70.63±4.1	72.71±3.9
	DKVMN	48.84±3.8	60.72±3.5	67.16±5.0	48.23±3.8	63.20±5.2	67.31±4.6	45.31±3.1	70.6±3.4	71.8±2.9	45.11±3.5	71.82±4.0	73.66±3.6
	SAKT	48.13±3.3	61.87±3.7	68.34±4.2	47.42±3.1	64.07±4.6	68.42±3.8	44.93±2.3	71.06±2.1	72.62±1.8	44.37±2.8	72.57±3.6	74.52±2.7
	AKT-NR	47.37±2.8	62.42±2.9	69.13±4.2	46.84±2.5	64.81±1.8	69.16±2.4	44.57±1.8	71.83±2.4	73.24±0.7	43.93±1.9	73.34±2.5	75.34±1.3
	GIKT	46.24±2.9	64.86±3.0	70.37±3.1	45.32±2.8	66.92±1.6	70.63±1.2	43.66±1.2	72.31±1.6	74.51±0.9	42.14±1.0	74.39±1.4	76.47±0.9
Pre-training based	PEBG+DKT	47.08±2.5	63.77±2.6	69.43±1.8	46.26±2.2	65.26±0.7	69.74±1.0	44.15±0.7	71.70±1.3	73.86±1.1	43.36±0.9	73.81±1.3	75.68±0.9
	PEBG+DKVMN	46.41±1.8	64.43±2.4	70.15±2.0	45.77±1.2	66.62±1.3	70.80±0.8	43.36±0.8	72.23±0.9	74.75±0.6	42.90±1.1	74.97±0.9	76.90±1.2
	BiCo+DKT	45.27±2.2	65.71±2.9	70.17±1.8	44.76±0.9	67.91±1.1	69.32±0.6	43.13±1.1	71.89±0.4	73.44±0.8	42.52±1.2	76.36±1.3	77.32±0.8
	BiCo+DKVMN	41.15±1.9	67.24±2.3	71.42±1.9	43.68±1.3	68.84±0.8	70.49±1.1	42.18±0.6	73.15±0.8	75.17±0.7	41.27±0.6	72.25±1.1	78.45±0.6
	BiCo+SAKT	44.84±2.0	66.18±1.2	70.53±0.8	44.51±0.9	68.13±0.6	69.71±0.4	42.77±0.1	72.33±0.7	73.70±0.7	42.18±0.9	76.69±0.5	77.84±0.9
	BiCo+AKT-NR	44.53±1.4	66.57±1.5	70.82±1.1	44.23±1.0	68.37±1.2	70.11±0.6	42.34±0.4	72.64±0.5	74.14±0.1	41.83±0.5	77.11±0.7	78.14±0.4
	BiCo+GIKT	43.61±1.3	68.11±0.4	72.49±0.7	43.16±1.3	71.42±0.5	72.87±0.5	41.50±0.3	73.92±0.3	75.95±0.3	39.81±0.2	78.32±0.4	79.28±0.2
	BiCoEE+GIKT	**43.24±0.9**	**68.50±0.6**	**72.71±0.8**	**42.87±0.4**	**71.75±0.2**	**73.24±0.1**	**41.32±0.2**	**74.50±0.4**	**76.28±0.3**	**39.40±0.1**	**78.94±0.2**	**79.64±0.3**

5.1.2 Baselines and Evaluation Metrics. To show the improvement of our method to the mainstream KT models and verify the effectiveness of our proposed method, we compare it with some baselines. Specifically, we first applied it on two kinds of KT models: state-oriented models (DKT, DKVMN, AKT, GIKT) and prediction-oriented models (SAKT). Then we compare it with the state-of-the-art question embedding method without text information PEBG. Here we do not include RKT [6] and GKT [4] mentioned in Section 2 because: 1) the implementation of RKT requires the text information of questions; 2) a question utilized in GKT solely contains a single concept. In order to ensure the fairness, all models were tuned to have the best performance.The details are illustrated as follows:

- *DKT* [7] is an implementation using a LSTM which leverages a series of one-hot representations of question and trains a unified network for predicting students' next interaction on certain question.
- *DKVMN* [9] uses a key-value memory network to store underlying concept representations and states.
- *SAKT* [5] is a recently proposed deep-learning method based on self-attention mechanism to capture the relation between interactions.
- *AKT-NR* [3] is an attention-based neural network model and attention weights are computed by a distance-aware exponential decay with a global decay rate, where we remove the Rasch model-based embeddings.
- *GIKT* [8] employs high-order question-concept relation graphs into question and concept representations for knowledge tracing.
- *PEBG* [24] formulates the question-concept relations as a bipartite graph and introduces a product layer to learn low-dimensional question embeddings for knowledge tracing.

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Moreover, we select evaluation metrics from both regression perspective and classification perspective. From the former perspective, we select Root Mean Square Error (RMSE, the lower the better) to quantify distances between predicted scores (i.e., continuous variable ranges from 0 to 1) and actual ones. From the latter perspective, we argue that students' response logs answered incorrectly or correctly can be represented as 0 and 1 respectively. Thereby, we select Accuracy (ACC, the higher the better) and Area Under the ROC Curve (AUC, the higher the better) for model evaluation.

5.2 Overall Performance (RQ1)

Table 3 and 4 illustrates the overall evaluation results about the student performance prediction task. We conduct 5-fold cross validation and report the average results. The best performance and the second best performance models are denoted in bold and underlined, respectively. From the result, we have the following observations:

- Almost all of the pre-training-based BiCo+ methods perform better than baselines at all data train/test split ratios, which may indicate that taking objective and subjective semantics into account for question embedding can achieve outstanding performance on the downstream KT task.
- BiCo+DKT and BiCo+DKVMN consistently perform better than PEBG+DKT and PEBG+DKVMN. This has two possible implications: (1) PEBG fails to take subjective students' response logs into consideration; (2) PEBG learns question embeddings and concept embeddings by constructing a simple folded bipartite graph, which may make two vertices connect with each other in case of insufficient or weak relations.
- Almost all BiCo+ methods are superior to baselines, probably because these baselines are compromised by the same issue, that is, data sparsity. On the contrary, BiCo incorporates the self-supervised learning mechanism so that it has the ability to be an antidote to data sparsity, which will be further discussed in Section 5.3.
- BiCoEE+GIKT performs the best, indicating that the loss enhancement can help our model learn more informative question embeddings to boost the performance of KT.

5.3 Ablation Study (RQ2)

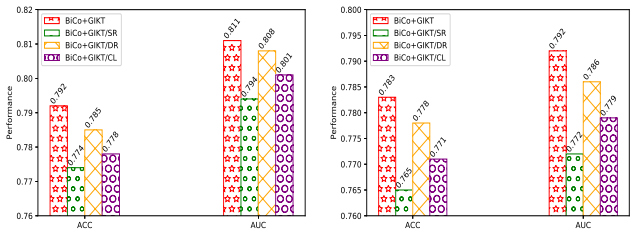
To get deep insights into BiCo, we conduct an ablation study to investigate the effectiveness of three important loss functions of our BiCo method on Assist09 and Junyi: (1) structure recovery loss; (2) difficulty recovery loss; (3) contrastive learning loss. We further set three comparative settings, and their performances have been shown in Fig. 4. The details of the three settings are listed below:

- BiCo+GIKT/SR does not consider the structure recovery loss, where the Eq. (13) is rewritten as $\min_{\theta} L = \min [L_{cl} + L_{dr}]$.
- BiCo+GIKT/DR does not consider the difficulty recovery loss, where the Eq. (13) is rewritten as $\min_{\theta} L = \min [L_{sr} + L_{cl}]$.
- BiCo+GIKT/CL does not consider the contrastive learning loss, where the Eq. (13) is rewritten as $\min_{\theta} L = \min [L_{sr} + L_{dr}]$.

To be fair, except for the changes mentioned above, the other parts of aforementioned methods and implementation settings remain identical. We report the results in Fig. 4 and find that:

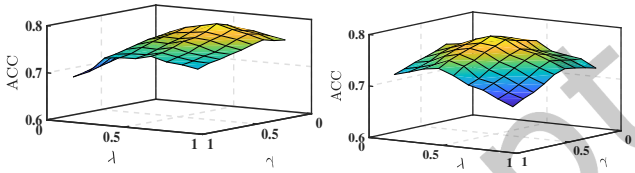
- BiCo+GIKT performs the best on both datasets, followed by BiCo+GIKT/DR, which indicates it is effective to take the question difficulty into modeling for the pre-training question embedding, but not the most important.
- The other two variants BiCo+GIKT/SR and BiCo+GIKT/CL do not perform better than BiCo+GIKT and BiCo+GIKT/DR on both datasets, which demonstrates the necessity of incorporating the structure recovery and contrastive loss. Moreover, BiCo+GIKT/SR performs the worst. This is probably because the structure

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(a) Assist09 (b) Junyi

Fig. 4. Performance comparison of ablation study on both datasets



(a) Assist09 (b) Junyi

Fig. 5. (a) and (b) show the ACC value achieved by BiCo with different λ and γ on Assist09 and Junyi, respectively.

recovery (i.e., those links between question nodes and concept nodes) can provide relatively objective information with less bias to our model. Although difficulty is a key property for a question, the question difficulty is a personalized as well as subjective definition for different students. Thus, taking the question difficulty into account leads to contributing less to our model than the structure recovery.

- Different datasets have different impacts on structure recovery and difficulty recovery. In Junyi, removing the difficulty recovery decreases the performance more significantly, which is contrary to Assist09. One reason for this trend is that the question-concept interaction information is pretty rich and difficulty labels in Assist09 are computed manually, however in Junyi, the question-concept interaction information is insufficient and those difficulty labels are manually annotated by domain experts.
- Dispensing with the contrastive learning loss, the performances also show a dramatically shrinking trend, which indicates that contrastive learning serves as an antidote to the aforementioned data sparsity issue indeed. We guess two possible reasons for this observation. First, the cross-view contrastive learning loss bridges both the Q2Q graph and the Q2C graph and provide more objective information to the former, since the Q2Q graph is manually computed so that there exists inevitable bias and sparsity (e.g., some question nodes are isolated). Second, we adopt rational strategies to select positive and negative samples, and these selected negative samples are hard enough to contribute enough information. By using these selected samples, the contrastive loss is task-relevant so that it can be conducted for refining encoders on the two views.

5.4 Sensitivity Evaluation (RQ3)

In this section, we investigate the effect of hyperparameters of BiCo with GIKT model according to the performance of the student response prediction task on both datasets. More specifically, we report the sensitivity analysis of

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three significant hyperparameters in BiCo, including the sparsity factor of the Q2Q graph (λ), the balance factor between two auxiliary tasks (γ) and the threshold of negatives and positives (k).

- **Analysis of** λ. The sparsity factor λ determines the number of edges in the Q2Q graph. We vary its value and corresponding results are shown in Fig. 5 (a) and (b). With the increase of λ, the performance on both datasets goes up first and then declines, and optimum point for Assist09 is at 0.4 and at 0.6 for Junyi. This has two possible implications. First, the number of response logs in Junyi is far more than Assist09, contributing more edges in the Q2Q graph. Thus, the value of λ in Junyi is bigger than Assist09 to provide more accurate and less redundant information. Second, there are same trends in both datasets, implying that λ with bigger value can generally encode more meaningful semantic information for question embeddings, whereas λ with too bigger value may lead to over-fitting. Hence, we set $\lambda = 0.4, 0.6$ in Assist09 and Junyi for obtaining the best results.
- **Analysis of** γ. The balance factor γ can be seen as a trade-off between the two auxiliary tasks (i.e., the contrastive learning loss and question difficulty recovery). Specifically, as γ increases, the performance of BiCo+GIKT firstly increase, but decrease when γ surpasses 0.6 and 0.5 in the corresponding datasets. This is beacuse BiCo prefers to mining self-supervised signals rather than difficulty information. In effect, the changing trends of performance in Fig. 5 (a) and (b) demonstrates the former statement indeed, it is more probable that the liabilities of difficulty information are limited in both datasets. However, we find that the difficulty information serves as vitamin to question representations in the two datasets, that is, even in small quantities could achieve comparable performance, while its deficiency would hinder the results severely. As a result, we set $\gamma = 0.6, 0.5$ in Assist09 and Junyi, because the performance of BiCo+GIKT achieves the best when it reaches the corresponding value.
- **Analysis of** k. The threshold k determines the number of positives and negatives during the process of co-contrastive learning. According to the results in Fig. 6, Assist09 is sensitive to k, and the best result is achieved when $k = 7$. However, Junyi behaves relatively stably, and the best result is achieved when $k = 15$.

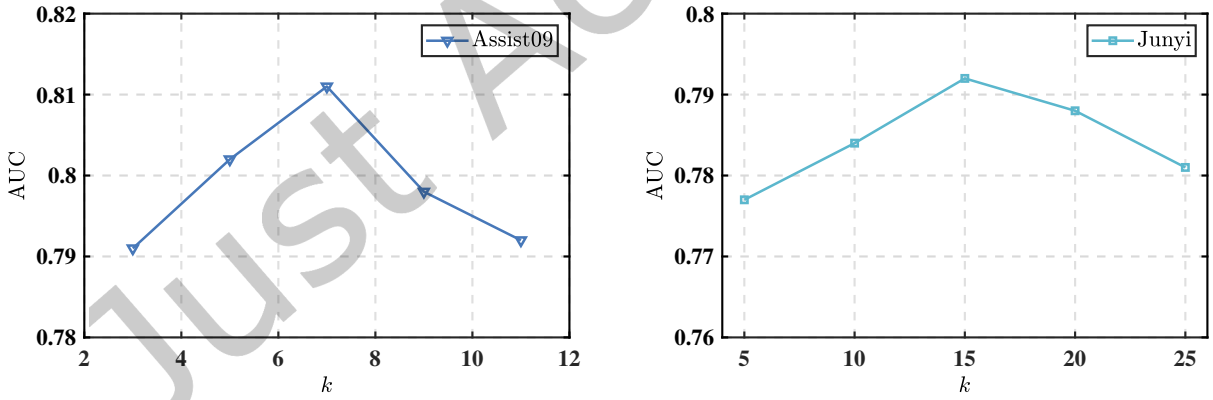


Fig. 6. Performance comparison with respect to k on both datasets

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5.5 Case Study (RQ4)

We perform a case study with a sampled example for the KT task with and without knowledge states of the student, to show the interpretation of question difficulty recovered by our proposed method (as illustrated in Fig. 7). From the results of Fig. 7, we can observe: (1) when predicting q_2 at timestep t_3, the result predicted by SAKT

is not consistent with the ground truth; (2) although this girl has answered the same question at timestep t_2 and t_3, the predicted value of SAKT has decreased from 44% to 35% and the one of DKVMN has increased from 48% to 61%; (3) this girl has never answered q_3 before, but both methods predict that she will fail to answer q_3.

In fact, for observation (1), since SAKT is the prediction-oriented method without knowledge states, it generates predictions heavily relying on the question embeddings. Along this line, this girl has left the wrong answer on q_2 at timestep t_2, so SAKT predicts that she cannot respond to the same question correctly at t_3. For observation (2), one possible reason for this shrinking trend is that the limited "wrong" label and the same question q_2 at timestep t_2 have the dominant impact on the predicted result of SAKT at timestep t_3. However, for DKVMN, the predicted results are determined by not only the "wrong" label and the same question but also her knowledge states which have evolved after some knowledge acquisition. Especially for observation (3), both methods show consistent results even if she has never answered q_3, which may indicate that the question embedding learned by our method covers the difficulty information effectively.

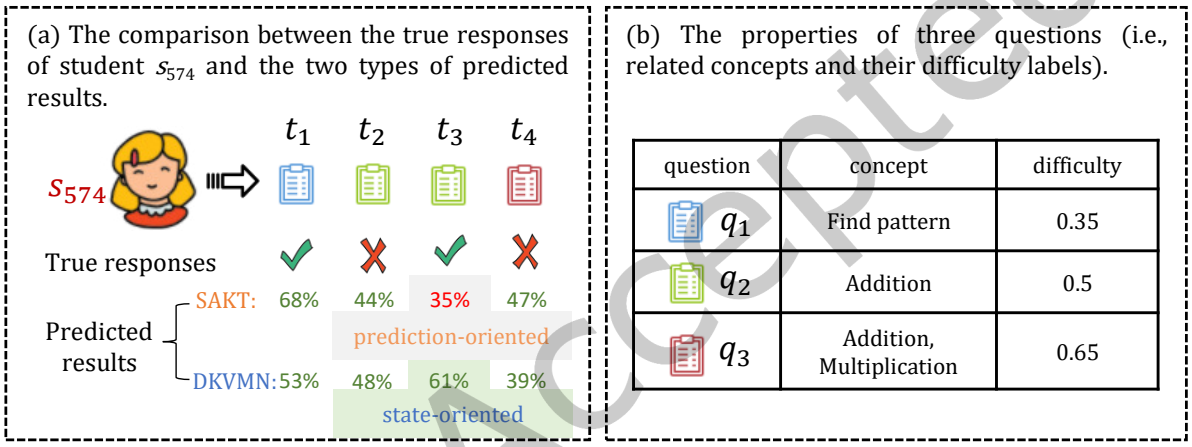


Fig. 7. A case study of the interpretation of question difficulty recovered by BiCo on Assist09. In this instance, a student with ID 574 has answered three questions at four different timesteps. We use SAKT and DKVMN to simulate the learning process of this girl, and obtain the different predicted results respectively.

6 CONCLUSIONS AND FUTURE WORK

In this paper, we present a novel BiCo method, focusing on improving knowledge tracing by capturing both the semantic and structural properties of questions in a self-supervised manner. Technically, on the basis of self-supervised learning paradigm, we first select two similar but distinct views (i.e., representing objective and subjective semantic perspectives) as the semantic source of question embeddings. Then, we design a primary task (structure recovery) together with two auxiliary tasks (question difficulty recovery and contrastive learning) to further enhance the representativeness of questions. Extensive experiments conducted on two real-world datasets show BiCo has higher expressive power that enables KT methods to effectively predict students' performance. To be specific, we have answered four research problems. In the first place, comparing with the state-of-the-art question embedding methods without text information, our method generally outperforms all the other methods on all datasets and all splits. Second, the performance results also demonstrate the effectiveness of taking students' response logs into account (i.e., Q2Q graph). Third, we investigate three hyperparameters' contributions to our model and find their best corresponding values on two datasets. Fourth, our case study with a sampled example

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for the KT task with and without knowledge states of the student show the interpretation of question difficulty recovered by our proposed method. Avenues of future work include: (1) generating harder negative samples for questions in the context of contrastive learning and (2) fusing more psychometric or signed network theories into question embedding methods to discriminate question embeddings with right or wrong responses.

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REFERENCES

[1] He, L., Tang, J., Li, X., Wang, P., Chen, F., Wang, T.: Multi-type factors representation learning for deep learning-based knowledge tracing. Word Wide Web Journal 2022, 1-30 (2022)
[2] Zhao, Y., Ma, H., Wang, W., Gao, W., Yang, F., He, X.: Exploiting multiple question factors for knowledge tracing. Expert Systems with Applications, 223, 119786 (2023)
[3] Ghosh, A., Heffernan, N., Lan, A.S.: Context-aware attentive knowledge tracing. In: Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. pp. 2330-2339. SIGKDD (2020)
[4] Nakagawa, H., Iwasawa, Y., Matsuo, Y.: Graph-based knowledge tracing: modeling student proficiency using graph neural network. In: Proceedings of 2019 WI International Conference on Web Intelligence. pp. 156-163. WI (2019)
[5] Pandey, S., Karypis, G.: A self attentive model for knowledge tracing. In: Proceedings of the 12th International Conference on Educational Data Mining. pp. 384-389. EDM (2019)
[6] Pandey, S., Srivastava, J.: RKT: Relation-aware self-attention for knowledge tracing. In: Proceeding of the 29th ACM International Conference on Information and Knowledge Management. pp. 1205-1214. CIKM (2020)
[7] Piech, C., Spencer, J., Huang, J., Ganguli, S., Sahami, M., Guibas, L., Sohl-Dickstein, J.: Deep knowledge tracing. In: Proceedings of the 28th International Conference on Neural Information Processing Systems. pp. 505-513. NeurIPS (2015)
[8] Yang, Y., Shen, J., Qu, Y., Liu, Y., Wang, K., Zhu, Y., Zhang, W., Yu, Y.: GIKT: A graph-based interaction model for knowledge tracing. In: European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases 2020. pp. 299-315. ECML-PKDD (2020)
[9] Zhang, J., Shi, X., King, I., Yeung, D.Y.: Dynamic key-value memory networks for knowledge tracing. Dynamic key-value memory networks for knowledge tracing. In: Proceedings of the 26th International Conference on World Wide Web. pp. 765-774. WWW (2017)
[10] Huo, H., Wong, D.F., Ni, L.M., Chao, L.S., Zhang, J.: Knowledge modeling via contextualized representations for LSTM-based personalized exercise recommendation. Information Sciences 523, 266-278 (2020)
[11] Wang, W., Ma, H., Zhao, Y., Yang, F., Chang, L.: Tracking knowledge proficiency of students with calibrated Q-matrix. Expert Systems With Applications 192, 116454 (2022)
[12] Huang, Z., Liu, X., Wang, H., Liu, Q., Chen, E., Ma, J., Su, Y., Tong, W.: DisenQNet: Disentangled representation learning for educational questions. In: Proceedings of the 27th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. pp. 696-704. SIGKDD (2021)
[13] Liu, Q., Huang, Z., Yin, Y., Chen, E., Xiong, H., Su, Y., Hu, G.: EKT: Exercise-aware knowledge tracing for student performance prediction. IEEE Transactions on Knowledge and Data Engineering 33(1), 100-115 (2021)
[14] Yin, Y., Liu, Q., Huang, Z., Chen, E., Tong, W., Wang, S., Su, Y.: QuesNet: A unified representation for heterogeneous test questions. In: Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. pp. 1328-1336. SIGKDD (2019)
[15] Liu, X., Zhang, F., Hou, Z., Mian, L., Wang, Z., Zhang, J., Tang, J.: Self-supervised learning: generative or contrastive. IEEE Transactions on Knowledge and Data Engineering, (2021)
[16] Chen, T., Kornblith, S., Norouzi, M., Hinton, G.: A simple framework for contrastive learning of visual representations. In: Proceedings of the 37th International Conference on Machine Learning. pp. 1597-1607. ICML (2020)
[17] He, K., Fan, H., Wu, Y., Xie, S., Girshick, R.: Momentum contrast for unsupervised visual representation learning. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 9729-9738. CVPR (2020)
[18] Hassani, K., Khasahmadi, A.H.: BERT: Pre-training of deep bidirectional transformers for language understanding. In: Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies. pp. 4171-4186. NAACL-HLT (2019)

Just Accepted

[19] Lan, Z., Chen, M., Goodman, S., Gimpel, K., Sharma, P., Soricut, R.: ALBERT: A lite BERT for self-supervised learning of language representations. In: Proceedings of the 8th International Conference on Learning Representations. pp. 4116-4126. ICLR (2020)
[20] Yu, J., Yin, H., Xia, X., Chen, T., Cui, L., Hung, N. Q. V. Are graph augmentations necessary? Simple graph contrastive learning for recommendation. In: Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval. SIGIR (2022)
[21] Wu, J., Wang, X., Feng, F., He, X., Chen, L., Lian, J., Xie, X.: Self-supervised graph learning for recommendation. In: Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval. pp. 726-735. SIGIR (2021)
[22] Wang, X., Liu, N., Han, H., Shi, C.: Self-supervised heterogeneous graph neural network with co-contrastive learning. In: Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery and Data Mining. pp. 1726-1736. SIGKDD (2021)
[23] Tang, J., Qu, M., Wang, M., Zhang, M., Yan, J., Mei, Q.: LINE: Large-scale information network embedding. In: Proceedings of the 24th International Conference on World Wide Web. pp. 1067-1077. WWW (2015)
[24] Liu, Y., Yang, Y., Chen, X., Shen, J., Zhang, H., Yu, Y.: Improving knowledge tracing via pre-training question embeddings. In: Proceedings of the 29th International Joint Conference on Artificial Intelligence. pp. 1577-1583. IJCAI (2020)
[25] Wang, W., Ma, H., Zhao, Y., Yang, F., Chang, L.: PERM: Pre-training question embeddings via relation map for improving knowledge tracing. In: Proceedings of the 27th Conference of the Database Systems for Advanced Applications. pp. 281-288. DASFAA (2022)
[26] Velickovic, P., Fedus, W., Hamilton, W.L., Lióa, P., Bengio, Y., Hjelm, R.D.: Deep graph infomax. In: Proceedings of the 7th International Conference on Learning Representations. ICLR (2019)
[27] Peng, Z., Huang, W., Luo, M., Zheng, Q., Rong, Y., Xu, T., Huang, J.: Graph representation learning via graphical mutual information maximization. In: Proceedings of the 29th International Conference on World Wide Web. pp. 259-270. WWW (2020)
[28] Park, C., Kim, D., Han, J., Yu, H.: Unsupervised attributed multiplex network embedding. In: Proceedings of the 34th AAAI Conference on Artificial Intelligence. pp. 5371-5378. AAAI (2020)
[29] Wang, W., Ma, H., Zhao, Y., Yang, F., Chang, L.: SEEP: Semantic-enhanced question embeddings pre-training for improving knowledge tracing. Information Sciences, 614, 153-169 (2022)
[30] Wang, C., Ma, W., Zhang, M., Lv, C., Wan, F., Lin, H., Tang, T., Liu, Y., Ma, S.: Temporal cross-effects in knowledge tracing. In: Proceedings of the 14th ACM International Conference on Web Search and Data Mining. pp. 517-525. WSDM (2021)
[31] Zhang, Q., Sun, Z., Hu, W., et al.: Multi-view knowledge graph embedding for entity alignment. In: Proceedings of the 28th International Joint Conference on Artificial Intelligence. pp. 5429-5435. IJCAI (2019)
[32] Hao, J., Chen, M., Yu, W., Sun, Y., Wang, W.: Universal representation learning of knowledge bases by jointly embedding instances and ontological concepts. In: Proceedings of the 25th ACM SIGKDD Conference on Knowledge Discovery and Data Mining. pp. 1709-1719. SIGKDD (2019)
[33] Hassani, K., Khasahmadi, A. H.: Contrastive multi-view representation learning on graphs. In: Proceedings of the 37th International Conference on Machine Learning. pp. 4116-4126. PMLR (2020)
[34] He, X., Deng, K., Wang, X., Li, Y., Zhang, Y., Wang, M.: LightGCN: Simplifying and powering graph convolution network for recommendation. In: Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval. pp. 639-648. SIGIR (2020)
[35] Aaron van den Oord, Yazhe Li, and Oriol Vinyals.: Representation learning with contrastive predictive coding. arXiv preprint arXiv:1807.03748 (2018).
[36] Bi, H., Ma, H., Huang, Z., Yin, Y., Liu, Q., Chen, E., Su, Y., Wang, S.: Quality Meets Diversity: A model-agnostic framework for computerized adaptive testing. In: Proceedings of the 20th IEEE International Conference on Data Mining. pp. 42-51. ICDM (2020)
[37] Xia, X., Yin, H., Yu, J., Shao, Y., Cui, L.: Self-supervised graph co-Training for session-based recommendation. In: Proceeding of the 30th ACM International Conference on Information and Knowledge Management. pp. 2180-2190. CIKM (2021)

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